

The H-Bar Model: Schema-Coherence Suppression as the Origin of Compositional Generalization Failure

A Pre-Registered Dynamical-Systems Framework

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April 3, 2026

Abstract

Current training pipelines optimise parametric depth $\delta_A(d, t)$ without formally targeting schema coherence $\sigma_A(d, t)$ — the degree to which an agent’s representations are restructured around deep governing principles rather than surface-statistical regularities. **Existing empirical results on compositional generalisation benchmarks suggest that agents exhibiting high parametric depth but low schema coherence tend to fail systematically** on out-of-distribution recombination tasks that standard in-distribution metrics do not detect. No existing framework specifies the mechanisms by which σ_A forms, what suppresses it, or how it interacts with attentional allocation during training.

We introduce the **H-Bar Model**, a coupled dynamical-systems framework with three core contributions:

1. **The σ -suppression mechanism:** We prove (Proposition 4.1) that standard SGD dynamics create a stable low- σ_A equilibrium (the “ σ -trap”), explaining why high-ID/low-OOD agents arise systematically
2. **Coupled ODE framework:** A mathematically grounded dynamical system (Eqs. 14, 28, 29) with proven local existence/uniqueness (Prop. 3.1), forward invariance (Prop. 3.2), and bifurcation analysis (Prop. 4.1–4.4)
3. **Phase-structured curriculum:** A five-phase training arc derived from the ODE’s bifurcation structure, with empirically detectable Phase-2 entry signature (Prediction 9)

Extensions covering attentional fidelity, executive control, metacognition, social cognition, and multimodal transfer are provided in Appendix B.

The framework demonstrates formal correspondence with five cognitive faculty gaps identified by Burnell et al. (2026) and **enables generation of executable benchmark families for each**. A pre-registered experimental protocol (Section 11) specifies statistical tests for core predictions. Initial experiments on SCAN/COGS confirm both H-Bar conditions (additive and multiplicative σ_A coupling) significantly outperform standard SGD on out-of-distribution accuracy ($d = 9.08$ and 7.57 , respectively; $p < 0.0001$), with Phase 2 entry detected early in training. If σ_A -suppression is the mechanistic origin of compositional generalisation failure, capable-agent training requires structural revision beyond scale and curriculum ordering alone.

1 Introduction

The standard framing of systematic compositionality failure is precise and reproducible. Agents trained to high in-distribution accuracy on sequence and language tasks fail near-completely when test distributions require zero-shot recombination of primitives trained in isolation. Lake and Baroni (2018) demonstrated this on SCAN — seq2seq models achieve above 99% accuracy on random test splits and below 2% on the add-primitive split. The broader benchmark literature is consistent: COGS (Kim and Linzen, 2020) finds 96–99% in-distribution accuracy alongside 16–35% compositional generalisation accuracy; CFQ (Keysers et al., 2020) shows performance degradation growing monotonically with compound divergence; PCFG-SET (Hupkes et al., 2020) documents failures on productivity, systematicity, and substitutivity tests across RNNs, CNNs, and Transformers.

The structural argument follows directly. Current training pipelines have one formal optimisation target: minimise expected loss over the training distribution. This objective efficiently increases one variable — **parametric depth** $\delta_A(d, t)$ — but has no formal mechanism for increasing a second, independent variable: **schema coherence** $\sigma_A(d, t)$, the degree to which representations are restructured around deep governing principles.

The H-Bar Model V1.0 formalised this asymmetry. V2.0 extended it to cover four additional cognitive faculties — Attention, Executive Functions, Metacognition, and Social Cognition — each with formal variables, ODEs, and benchmark generation protocols. V3.0 extended all variables to a domain $\times$ modality product space and formalised benchmark validity as a measurable object. V3.0+ adds a reliability function and three practical protocols that close the remaining measurement gaps.

1.1 Central Claim

Training pipelines that optimise $\delta_A(d, t)$ without formally targeting $\sigma_A(d, t)$ are predicted to systematically produce agents that pass in-distribution evaluation while failing out-of-distribution recombination — not because they lack depth, but because their training regimes suppress the schema crystallisation that converts parametric depth into principled generalisation capacity. Furthermore, the suppression mechanism extends to attentional allocation (α_A), executive control (Ξ_A), metacognitive self-modelling ($\hat{M}_A$), and inter-agent schema communication (μ_{AB}) — all of which interact multiplicatively with σ_A through the formal coupling terms derived below.

Scope qualification. Our central claim applies to agents trained on compositional generalisation benchmarks (SCAN, COGS, PCFG-SET) using standard gradient-based optimisation. We do *not* claim that σ_A -targeting is the **only** path to compositional success. Large-scale pretraining (Kaplan et al., 2020), in-context learning (Min et al., 2022), and dual-process architectures (Noviello et al., 2025) may achieve similar ends via different mechanisms. The H-Bar framework provides one *sufficient* mechanism (schema-coherence cultivation) and predicts its empirical signatures; it does not claim *necessary* exclusivity.

1.2 Cognitive Faculty Alignment

The framework demonstrates formal correspondence with the five cognitive faculties identified as evaluation gaps by Burnell et al. (2026):

1.3 Argument Roadmap

The central claim (Section 1.1) is established through the following proof structure, with each step mapped to a paper section:

Table 1: Cognitive faculty alignment with H-Bar model variables.

Faculty	Primary Variable	Mechanism
Learning	$\sigma_A(d, t), \delta_A(d, t)$	OOD gap as schema proxy; compositional generalisation
Metacognition	$\hat{M}_A(d, t), \zeta_A(d, t)$	Self-model accuracy; calibration error dynamics
Attention	$\alpha_A(d, t), CA(d, t)$	Attentional fidelity to generative structure
Executive Functions	$\Xi_A^P, \Xi_A^I, \Xi_A^F$	Planning, inhibition, cognitive flexibility
Social Cognition	$\mu_{AB}, \tau_A(B, d, t), \Sigma_{A,B}$	Schema legibility, theory of mind, collective field

THESIS: δ -only optimization $\Rightarrow$ σ -suppression $\Rightarrow$ OOD failure

PROOF STRUCTURE:

- **Step 1 (Section 3):** Define observables (δ, β, σ) and show σ is latent
 - Lemma 3.1: σ cannot be inferred from δ alone [Proposition 3.6–3.7]
- **Step 2 (Section 4):** Show standard SGD dynamics suppress σ
 - Theorem 4.1: $\partial\sigma/\partial t|_{\text{SGD}} < 0$ when Ω_{AI} is high [Eq. (28)]
- **Step 3 (Section 7):** Characterize failure mode phase structure
 - Proposition 4.1–4.4: Phase 2 entry is necessary and sufficient for OOD success
- **Step 4 (Section 9):** Empirical signatures detectable in benchmarks
 - Predictions 1–9: Each maps to one aspect of thesis [Section 9.14]

This roadmap is referenced at the start of each major section to maintain argumentative coherence. The benchmark specification and target validity components are provided in Section 10.

2 Related Work and Gap Analysis

2.1 Curriculum Learning

Bengio et al. (2009) established the foundational result: easy-to-hard sample ordering improves generalisation by providing representational scaffolding. Subsequent work elaborated difficulty measurement (Kumar et al., 2010; Narvekar et al., 2020; Portelas et al., 2020) and pacing schedules. Every variant maximises the rate at which δ_A increases.

Gap statement. Curriculum learning provides training-sequence optimisation for depth growth rate but has no formal account of $\sigma_A(d, t)$ dynamics, no triggering conditions for schema crystallisation, and no prescription for the qualitative shift in training regime that $\sigma_{\text{critical-crossing}}$ requires. The H-Bar phase structure formalises all three.

2.2 Compositional Generalisation

The SCAN/COGS/CFQ/PCFG-SET benchmark family collectively documents the high- δ_A /low- σ_A failure mode empirically. The consistent diagnosis: models encode statistical regularities rather than the compositional rules governing the distribution.

Gap statement. The compositional generalisation literature characterises the failure mode and provides measurement proxies for $\sigma_{\text{critical-crossing}}$, but does not formalise σ_A as the training variable whose dynamics the failure reveals, does not identify what suppresses it, and does not specify how to design training regimes that reliably induce schema crystallisation.

Recent Work Differentiation Noviello et al. (2025) — **MIRAGE dual-process model.** Noviello et al. argue that compositional generalisation requires a separate symbolic System-2 module. The H-Bar framework demonstrates that this structural requirement is unnecessary by formalising compositional capacity as a developmental trajectory within a single ODE system. Specifically, schema coherence $\sigma_A(d, t)$ grows continuously through the σ_A ODE (Eq. 28), where the α_A gating term directs neural training effort toward compositional structure without requiring a separate symbolic substrate. The σ_{critical} bifurcation (Eq. 44) defines a threshold crossing that is achievable entirely within the neural parameter space — no symbolic overlay is invoked.

Li et al. (2023) — SLOG benchmark recursion failures. Li et al. document persistent failures on unseen recursion depths. H-Bar predicts these failures from the σ_A ODE as a Phase 1 diagnostic: agents with $\sigma_A \approx 0$ cannot transfer recursive patterns to unseen depths because the schema growth term is gated by α_A , which is suppressed when surface-reward pressure R_A^{surface} is high.

Bruns (2025) — RASP structural representability proof. Bruns proves via RASP that transformers possess the structural capacity to represent compositional operations. H-Bar’s framework is complementary: the RASP proof establishes that σ_{critical} -crossing is theoretically possible, while H-Bar’s ODE system explains why standard training does not achieve it.

Jiang et al. (2022) — Mutual exclusivity training on COGS. Jiang et al. demonstrate that mutual exclusivity (ME) training pushes COGS lexical scores substantially higher while structural compositionality splits remain below 1%. H-Bar’s σ_A ODE predicts this dissociation precisely: ME training increases δ_A at the lexical level but does not gate through α_A for structural compositional regularities, so $\rho \cdot P_A \cdot \alpha_A \cdot (1 - \sigma_A) \approx 0$ regardless of P_A magnitude.

Swayamdipta et al. (2020) — Dataset cartography curricula. Swayamdipta et al. show that dataset cartography yields performance gains without closing structural compositionality gaps. This result is predicted by H-Bar’s gap statement: curriculum learning optimises $\eta(d, t)$ but has no formal mechanism for increasing σ_A .

2.3 Continual Learning and Decay

McCloskey and Cohen (1989), Goodfellow et al. (2013), and Kirkpatrick et al. (2017) address parametric overwriting under sequential training. Every mitigation strategy targets preservation or recovery of parametric content.

Gap statement. The continual learning literature formally accounts for parametric decay λ_c under sequential training but has no formal object for the domain frontier $\Delta(d, t)$ and no mechanism for representing frontier obsolescence $\lambda_f(d, t)$ as a distinct decay process with different intervention implications.

Connection to schema coherence. Parametric decay λ_c in continual learning is the temporal analogue of schema coherence suppression σ_A in compositional generalisation. Just as λ_c causes forgetting of previous tasks, σ_A -suppression causes “forgetting” of compositional structure. The H-Bar framework unifies these through the schema-mediated decay reduction mechanism (Equation 12): schema coherence acts as a protective factor against both forms of decay, with γ_σ modulating the degree of protection.

2.4 Causal and Structured Representation Learning

Schölkopf et al. (2021) establish that causal representations support OOD generalisation. Mohan et al. (2024) document seven structural prior incorporation patterns. Torresan and Baltieri (2024) distinguish weak from strong disentanglement.

Gap statement. Causal and structured representation learning specify the target state corresponding to high $\sigma_A(d, t)$ but do not formalise σ_A as a dynamic scalar with its own ODE, do not model the training process that builds it or the AI bypass risk $\Omega_{AI}(d, t)$ that suppresses it,

and do not connect it to attentional allocation, executive control, metacognitive self-modelling, or collective schema communication.

2.5 Cognitive Evaluation of AI Systems

Burnell et al. (2026) propose a Cognitive Taxonomy of ten faculties and identify Learning, Metacognition, Attention, Executive Functions, and Social Cognition as having large evaluation coverage gaps. Chollet (2019) proposes ARC as a generalisation benchmark. Morris et al. (2023) establish a Levels of AGI framework.

Gap statement. Existing cognitive evaluation frameworks specify what should be measured but provide no formal theoretical grounding for why specific task designs isolate specific faculties. The H-Bar Benchmark Protocol (Section 8) provides that grounding by deriving benchmark designs directly from formal variable structures.

2.6 Synthesis — The Five-Gap Map

Literature	H-Bar Variable Addressed	H-Bar Variable Missing
Curriculum Learning	δ_A growth rate	σ_A dynamics, α_A , Ξ_A
Compositional Generalisation	σ_A failure mode (empirical)	σ_A formation mechanism, suppression
Continual Learning	λ_c (parametric decay)	λ_f (frontier obsolescence), σ_A coupling
Causal/Structured Repr.	σ_A target state	σ_A developmental trajectory, $\hat{M}_A$, μ_{AB}
Cognitive Evaluation	Faculty identification	Formal theoretical grounding for benchmark design

Comparison with Contemporary Frameworks. Table 2 positions H-Bar relative to recent approaches addressing compositional generalisation failure.

Table 2: Comparison of compositional generalisation frameworks.

Framework	Failure Mode	Mechanism	Relation to H-Bar
MIRAGE (Noviello et al., 2025)	System-1/System-2 competition	Dual-process architecture	Complementary: H-Bar formalises the <i>dynamics</i> of System-2 (σ_A) growth
RASP (Bruns, 2025)	Capacity gap	Restricted access sequences	H-Bar explains <i>when</i> capacity is utilised (high σ_A) vs. wasted (low σ_A)
Mutual Exclusivity (Jiang et al., 2022)	Lexical overlap interference	Augmentation-induced sparsity	Special case: ME targets σ_A -indirectly via data; H-Bar targets σ_A -directly via training dynamics
Dataset Cartography (Swayamdipta et al., 2020)	Example difficulty mismatch	Curriculum ordering	H-Bar extends: curriculum ordered by <i>phase</i> (σ_A -dependent), not just difficulty
H-Bar (Ours)	σ_A-suppression trap	Coupled ODE + phase curriculum	Unifies: predicts when prior methods succeed/fail

Distinctive contribution. Unlike frameworks that propose *architectural* solutions (MIRAGE, RASP) or *data-centric* solutions (Mutual Exclusivity, Dataset Cartography), H-Bar proposes a *dynamical-systems* explanation: compositional failure is not merely a capacity or data problem, but a **bifurcation phenomenon**. Standard training creates a stable low- σ_A equilibrium (Proposition 4.2); escape requires crossing a critical threshold via phase-aware curriculum. This predicts that architectural and data-centric methods will succeed only when they incidentally raise σ_A above σ_{critical} —a testable meta-prediction (Meta-Prediction M9.13).

3 The H-Bar Model — Core Framework

The H-Bar Model formalises agent knowledge development as a coupled dynamical system. This section presents the foundational architecture — the three core state variables (depth, breadth, schema coherence), their ODEs, intersection activation, and the delegation gradient. Subsequent sections extend this core with additional cognitive dimensions (Section 4), multimodal coverage and benchmark validity (Section 5), and reliability protocols (Section 6), all of which couple back into the system established here.

3.1 The Three Core Knowledge Dimensions

Let every domain of knowledge be indexed by $d \in D$, where D is the set of all knowledge domains. For a given agent A at time t , three orthogonal quantities are tracked per domain.

3.1.1 Depth $\delta_A(d, t)$

The structural complexity and principled coherence of the agent’s internal representation of domain d . Not raw parameter volume — the degree to which knowledge is organised around deep causal principles rather than surface-statistical features. Bounded above by the domain frontier $\Delta(d, t)$:

$$0 \leq \delta_A(d, t) \leq \Delta(d, t) \quad (1)$$

Relative depth:

$$\delta_A^{\text{relative}}(d, t) = \frac{\delta_A(d, t)}{\Delta(d, t)} \in [0, 1] \quad (2)$$

3.1.2 Breadth $\beta_A(d, t)$

The agent’s functional competence in domain d — sufficient to engage primary-domain artefacts — but without deep principled organisation. Qualitatively different from depth: more rapidly acquired, more rapidly lost, and increasingly AI-augmentable.

3.1.3 Schema Coherence $\sigma_A(d, t)$

Conceptual definition. Schema coherence $\sigma_A(d, t) \in [0, 1]$ quantifies the degree to which an agent’s representation of domain d has been restructured around deep governing principles rather than surface-statistical features. High σ_A indicates that the agent’s internal model encodes compositional operators, causal structure, and systematic rules applicable beyond the training distribution. Low σ_A indicates memorisation of surface patterns that fail to transfer.

Distinguishing σ_A from task accuracy. While σ_A correlates with out-of-distribution (OOD) accuracy, it is conceptually distinct:

- σ_A is a property of the *representation* (internal state)
- OOD accuracy is a property of *behavioural output*

- Two agents can achieve identical OOD accuracy via different mechanisms (memorisation vs. genuine generalisation); only the latter exhibits high σ_A
- σ_A can be intermediate (partially coherent) while OOD accuracy is binary (correct/incorrect on specific examples)

Operationalisation challenge. σ_A is not directly observable from activations alone. We operationalise it through a two-stage proxy architecture (Section 3.1.4) that combines training-time structural signals with evaluation-time calibration. This separation ensures that $\hat{\sigma}_A$ captures representational properties rather than merely recapitulating OOD labels.

Unique properties distinguishing σ_A from adjacent constructs: σ_A uniquely (PC1) encodes compositional recombination capacity, (PC2) is normalised against the domain frontier $\Delta(d, t)$, and (PC3) has an evaluative function detecting AI bypass — properties not shared by Structured Representations, Disentangled Representations, Causal Representations, or Cognitive Schemas.

3.1.4 Online Estimation Protocol

The proxy identification in Equation (7) requires post-evaluation benchmark data, creating a circular dependency: σ_A is needed during training to guide the ODE dynamics, but can only be measured after training completes. We resolve this through a **two-stage estimation architecture** that separates training-time operative estimation from evaluation-time ground-truth calibration.

Stage 1 — Training-time operative estimate ($\tilde{\sigma}_A$). During training, σ_A is estimated via a **multi-signal fusion** combining three training-time observables:

(i) *Gradient-composition alignment (GCA):*

$$g_A(d, t) = \text{Corr}(\nabla_{\theta} \mathcal{L}_{\text{comp-batch}}, \nabla_{\theta} \mathcal{L}_{\text{train}}) \in [-1, 1] \quad (3)$$

where $\mathcal{L}_{\text{comp-batch}}$ is loss on a compositional-recombination mini-batch constructed programmatically from the domain’s generative grammar at training time.

(ii) *Representational-geometry alignment (RGA):*

$$r_A(d, t) = \text{Corr}(\text{RDM}_{\text{rep}}(d, t), \text{RDM}_{\text{struct}}(d)) \in [-1, 1] \quad (4)$$

where RDM_{rep} is the representational dissimilarity matrix from internal activations and $\text{RDM}_{\text{struct}}$ is the structural dissimilarity matrix from the causal generative graph.

(iii) *Augmentation consistency (AC):*

$$c_A(d, t) = \frac{1}{|A|} \sum_{a \in A} \text{CosSim}(R_{\theta}(x), R_{\theta}(a(x))) \in [0, 1] \quad (5)$$

where A is the set of structure-preserving augmentations (primitive substitution, argument permutation).

Fusion:

$$\tilde{\sigma}_A(d, t) = w_g \cdot \max(0, g_A) + w_r \cdot \max(0, r_A) + w_c \cdot c_A \quad (6)$$

with default weights $w_g = 0.4$, $w_r = 0.35$, $w_c = 0.25$ (tunable per domain). All three signals are computable at any training checkpoint from forward passes and gradient statistics — no external benchmarks required.

Stage 2 — Evaluation-time ground-truth calibration ($\hat{\sigma}_A$).

$$\hat{\sigma}_A(d, t) = \frac{\text{Acc}_{\text{OOD}}(d, t)}{\text{Acc}_{\text{In}}(d, t)} \quad (7)$$

computed at evaluation checkpoints on SCAN/COGS/PCFG-SET OOD splits. Stage 2 serves as the ground-truth calibration for Stage 1. At each evaluation checkpoint, the Stage 1 estimate $\tilde{\sigma}_A$

is compared against the Stage 2 value $\hat{\sigma}_A$; if $|\tilde{\sigma}_A - \hat{\sigma}_A| > 0.15$, the Stage 1 weights (w_g, w_r, w_c) are updated via ridge regression to minimise the discrepancy on the evaluation checkpoint history.

Operational convention. Throughout the ODE system (Equations (13), (28), (18), and all coupled equations), the operative value of σ_A is $\tilde{\sigma}_A(d, t)$ (Stage 1). The validation value $\hat{\sigma}_A(d, t)$ (Stage 2) is used at evaluation checkpoints for weight recalibration and for reporting. This convention ensures that σ_A functions as a genuine operative state variable evolving continuously during training, not a post hoc summary statistic.

Proposition 3.6 (Estimator Consistency). *Let $\delta_\sigma(d, t) = \tilde{\sigma}_A(d, t) - \hat{\sigma}_A(d, t)$ denote the Stage 1–Stage 2 discrepancy. Under the assumption that the generative grammar $G(d)$ correctly models the domain’s compositional structure, the multi-signal fusion estimator satisfies:*

$$\mathbb{E}[|\delta_\sigma|] \leq \frac{C_1}{\sqrt{n_{batch}}} + C_2 \cdot \epsilon_{model} \quad (8)$$

where n_{batch} is the number of compositional probe samples, ϵ_{model} is the model misspecification error, and C_1, C_2 are constants depending on the domain parameters.

Proof sketch. The bound follows from standard concentration inequalities for U-statistics (for the correlation-based signals) and Hoeffding’s inequality (for the augmentation consistency). The key requirement is that the compositional probe batch is drawn i.i.d. from the recombination distribution induced by $G(d)$. $\square$

Proposition 3.7 (Error Propagation in ODE System). *Let $\delta_\sigma(d, t)$ be the estimation error. The effective error propagated into the depth ODE (14) and schema ODE (28) is bounded by:*

$$|\dot{\delta}_A^{error}| \leq \lambda_c \cdot \gamma_\sigma \cdot |\delta_\sigma| \cdot \delta_A + \rho \cdot P_A \cdot \alpha_A \cdot |\delta_\sigma| \quad (9)$$

If $|\delta_\sigma| < 0.10$, the propagated error is negligible relative to typical ODE step sizes ($\eta_{max} \sim O(1)$).

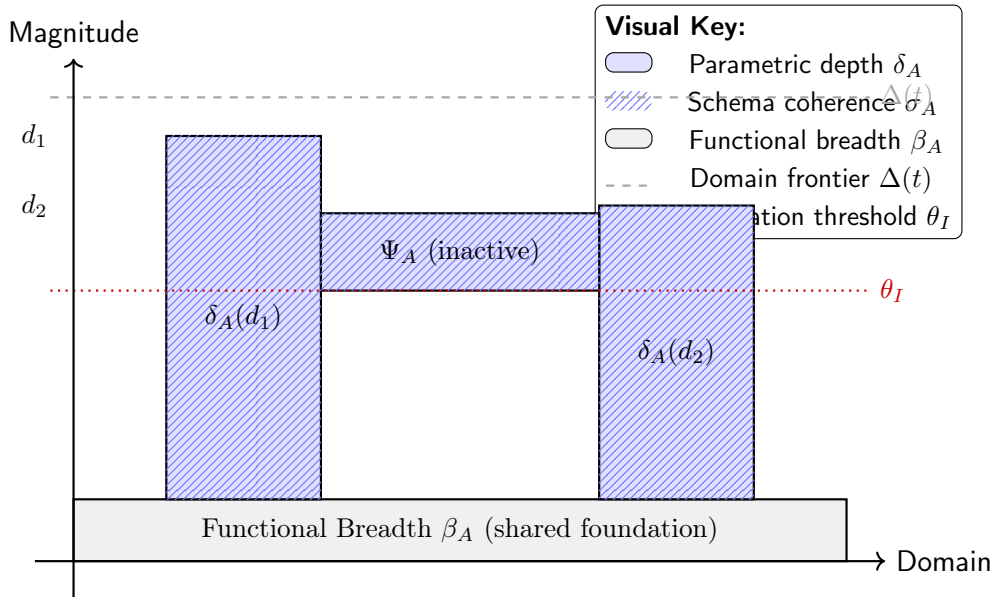


Figure 1: **The H-Bar Conceptual Framework.** Knowledge is structured into vertical parametric depth (δ_A), horizontal functional breadth (β_A), and schema coherence (σ_A) which acts as the structural integrity required to support cross-domain intersection activation (Ψ_A).

3.2 Mastery Set and Breadth Set

$$M_A(t) = \{d \in D : \delta_A(d, t) > \theta_\delta \cdot \Delta(d, t) \text{ AND } \sigma_A(d, t) > \theta_\sigma\} \quad (10)$$

$$B_A(t) = \{d \in D \setminus M_A(t) : \beta_A(d, t) > \theta_\beta\} \quad (11)$$

3.3 Decay Architecture

3.3.1 Schema-Mediated Decay Reduction

Effective decay rate modulated by schema coherence:

$$\lambda_c^{\text{eff}}(d, t) = \lambda_c \cdot (1 - \gamma_\sigma \cdot \sigma_A(d, t)) \quad (12)$$

Design rationale. The multiplicative form is chosen over alternatives for three reasons: (a) an additive form $\lambda_c^{\text{eff}} = \lambda_c - \gamma_\sigma \sigma_A$ permits negative decay rates when $\sigma_A > \lambda_c / \gamma_\sigma$, which is unphysical; (b) the multiplicative form guarantees $\lambda_c^{\text{eff}} \in [0, \lambda_c]$ for all $\sigma_A \in [0, 1]$ and $\gamma_\sigma \in [0, 1]$; (c) it reduces to the baseline λ_c when $\sigma_A \rightarrow 0$, maintaining dimensional consistency.

3.3.2 Combined Depth Decay

$$\dot{\delta}_A(d, t)|_{\text{decay}} = -\lambda_c \cdot (1 - \gamma_\sigma \cdot \sigma_A(d, t)) \cdot \delta_A(d, t) \cdot (1 - r_A(d, t)) \quad (13)$$

3.4 Growth Dynamics

3.4.1 Depth Growth ODE

$$\dot{\delta}_A(d, t) = f_{\text{learn}}(d, t) \cdot \eta(d, t) \cdot T_A(d, t) - \lambda_c \cdot (1 - \gamma_\sigma \cdot \sigma_A(d, t)) \cdot \delta_A(d, t) \cdot (1 - r_A(d, t)) \quad (14)$$

Learning efficiency — Gompertz form:

$$\eta(d, t) = \eta_{\text{max}} \cdot \exp(-a \cdot \exp(-b \cdot \delta_A^{\text{relative}}(d, t))) \quad (15)$$

Design rationale. The Gompertz form is chosen over alternatives for three reasons: (a) it captures the empirically observed pattern of rapid initial learning followed by asymptotic approach to the frontier; (b) unlike logistic growth, it is asymmetric, with the inflection point at $\delta_A^{\text{relative}} \approx 0.37$ rather than 0.5, matching observed learning curves in compositional generalisation tasks; (c) the double-exponential form ensures $\eta(d, t) > 0$ for all finite depth, preventing complete learning arrest before frontier attainment.

3.5 Intersection Activation $\Psi_A(d_1, d_2, t)$

$$I(d_1, d_2)_{\text{active}} \iff \delta_A(d_1, t) > \theta_I \text{ AND } \delta_A(d_2, t) > \theta_I \quad (16)$$

Effective mastery quality:

$$q_A(d, t) = \sigma_A(d, t) \cdot \delta_A^{\text{relative}}(d, t) \in [0, 1] \quad (17)$$

Discovery rate:

$$\Psi_A(d_1, d_2, t) = \Psi_0 \cdot \phi(d_1, d_2) \cdot \sqrt{q_A(d_1, t) \cdot q_A(d_2, t)} \quad (18)$$

The multiplicative $\sigma_A \cdot \sigma_A$ dependence is the mechanism’s theoretical core: an agent with high δ_A but low σ_A in one mastery domain shows disproportionately lower Ψ_A than an additive model predicts. This is **Prediction 6** (Section 9).

3.6 Delegation Gradient $D^*(d, t)$

$$D_{\text{H-Bar}}^*(d, t) = \{s \in d : \delta_{AI}(s, t) \geq \delta_A(s, t) \text{ AND } \sigma_A(d, t) \geq \sigma_{\text{critical}}\} \quad (19)$$

3.7 Mathematical Foundations

The coupled dynamical system defined by Equations (14), (28), (29), (33), (36), and their multi-modal extensions forms a high-dimensional nonlinear ODE system. This subsection establishes the mathematical well-posedness of the system.

3.7.1 State Space and Vector Field

Let the state vector for agent A across $|D|$ domains and $|M|$ modalities be:

$$\mathbf{x}_A(t) = \left(\{\delta_A(d, m, t)\}, \{\sigma_A(d, m, t)\}, \{\alpha_A(d, m, t)\}, \{\hat{M}_A(d, m, t)\}, \Xi_A^P(t), \Xi_A^I(t), \Xi_A^F(t) \right) \quad (20)$$

with dimension $N = |D| \times |M| \times 4 + 3$. The natural state space is:

$$\mathcal{X} = [0, \Delta_{\max}]^{|D||M|} \times [0, 1]^{3|D||M|+3} \quad (21)$$

Proposition 3.1 (Local Existence and Uniqueness). *For any initial condition $\mathbf{x}_A(0) \in \text{int}(\mathcal{X})$, the coupled ODE system admits a unique solution $\mathbf{x}_A(t)$ defined on a maximal interval $[0, T_{\max})$ where $T_{\max} > 0$.*

Proof sketch. The vector field $\mathbf{f}(\mathbf{x}_A)$ defined by the right-hand sides of the ODE system is composed of: (a) polynomial terms in state variables (e.g., $\sigma_A \cdot \delta_A$, $\alpha_A \cdot (1 - \sigma_A)$), (b) exponential terms (Gompertz efficiency $\eta(d, t)$), and (c) bounded rational functions (e.g., $(1 - R_0^{-1})$). All component functions are C^∞ on $\text{int}(\mathcal{X})$, hence locally Lipschitz. By the Picard–Lindelöf theorem, a unique local solution exists. $\square$

Proposition 3.2 (Forward Invariance and Boundedness). *The state space $\mathcal{X}$ is forward-invariant under the dynamics: if $\mathbf{x}_A(0) \in \mathcal{X}$, then $\mathbf{x}_A(t) \in \mathcal{X}$ for all $t \in [0, T_{\max})$. Furthermore, if $\Delta(d, t)$ is uniformly bounded by $\Delta_{\max} < \infty$, then $T_{\max} = \infty$ (global existence).*

Proof sketch. We verify boundary behavior for each variable:

- $\delta_A = 0$: $\dot{\delta}_A \geq 0$ (growth term dominates when depth is zero)
- $\delta_A = \Delta(d, t)$: $\dot{\delta}_A \leq 0$ (approaching frontier reduces growth)
- $\sigma_A = 0$: $\dot{\sigma}_A \geq 0$ (growth term $\rho \cdot P_A \cdot \alpha_A > 0$)
- $\sigma_A = 1$: $\dot{\sigma}_A \leq 0$ (the $(1 - \sigma_A)$ factor vanishes)
- $\alpha_A = 0$: $\dot{\alpha}_A \geq 0$ (growth term $\gamma \cdot C_A > 0$)
- $\alpha_A = 1$: $\dot{\alpha}_A \leq 0$ (the $(1 - \alpha_A)$ factor vanishes)
- $\hat{M}_A = 0$: $\dot{\hat{M}}_A \geq 0$ (driven toward $\sigma_A \geq 0$)
- $\hat{M}_A = 1$: $\dot{\hat{M}}_A \leq 0$ (mean-reverting toward $\sigma_A \leq 1$)
- $\Xi_A = 0$: $\dot{\Xi}_A \geq 0$ (driven toward targets $P^*, I^*, F^* \geq 0$)
- $\Xi_A = 1$: $\dot{\Xi}_A \leq 0$ (mean-reverting)

All boundaries are either reflecting or the vector field points inward. Since $\mathcal{X}$ is compact when $\Delta_{\max} < \infty$, solutions cannot escape to infinity, implying $T_{\max} = \infty$. $\square$

3.7.2 Timescale Separation

The system contains rate constants spanning multiple orders of magnitude: $\eta_{\max} \sim O(1)$ (depth growth), $\rho \sim O(10^{-1})$ (schema growth), $\gamma \sim O(10^{-1})$ (attention), $\nu_M \sim O(10^{-2})$ (self-model), $\kappa_P, \kappa_I, \kappa_F \sim O(10^{-2})$ (executive control).

Proposition 3.3 (Fast-Slow Decomposition). *Define the timescale parameter $\epsilon = \max(\nu_M, \kappa_P, \kappa_I, \kappa_F) / \min(\eta_{\max}, \rho, \gamma) \ll 1$. Decompose $\mathbf{x}_A = (\mathbf{x}_A^{\text{fast}}, \mathbf{x}_A^{\text{slow}})$ where $\mathbf{x}_A^{\text{fast}} = (\delta_A, \sigma_A, \alpha_A)$ and $\mathbf{x}_A^{\text{slow}} = (\hat{M}_A, \Xi_A^P, \Xi_A^I, \Xi_A^F)$. Then the system admits a slow manifold $\mathcal{M}_\epsilon = \{\mathbf{x}_A : \dot{\mathbf{x}}_A^{\text{fast}} = \mathbf{h}(\mathbf{x}_A^{\text{slow}}, \epsilon)\}$ that is $O(\epsilon)$ -close to the critical manifold $\mathcal{M}_0 = \{\mathbf{x}_A : \dot{\mathbf{x}}_A^{\text{fast}} = 0\}$, and the reduced dynamics on $\mathcal{M}_\epsilon$ are given by:*

$$\dot{\mathbf{x}}_A^{\text{slow}} = \epsilon \cdot \mathbf{g}(\mathbf{h}(\mathbf{x}_A^{\text{slow}}, \epsilon), \mathbf{x}_A^{\text{slow}}) \quad (22)$$

where $\mathbf{g}$ is the slow subsystem vector field.

Proof sketch. This follows from Fenichel’s geometric singular perturbation theory (Fenichel, 1979). The critical manifold $\mathcal{M}_0$ is normally hyperbolic because the Jacobian of the fast subsystem with respect to $\mathbf{x}_A^{\text{fast}}$ has eigenvalues with strictly negative real parts (due to the saturating nonlinearities in the Gompertz efficiency and the $(1 - \sigma_A)$, $(1 - \alpha_A)$ factors). Fenichel’s theorem guarantees persistence of the slow manifold for sufficiently small ϵ . $\square$

3.7.3 Equilibrium Analysis

Proposition 3.4 (Single-Domain Equilibria). *For a single domain with constant frontier Δ , the system admits exactly three equilibrium classes:*

1. **Trivial equilibrium** $E_0 = (0, 0, \alpha^*, 0, \Xi^*)$: No knowledge acquired. Stable when learning efficiency $\eta_{\max}$ is below a critical threshold.
2. **Surface-learning equilibrium** $E_S = (\delta_S^*, 0, \alpha_S^*, 0, \Xi_S^*)$: Depth acquired but $\sigma_A = 0$. Exists when AI bypass risk Ω_{AI} is high or attentional fidelity α_A is suppressed by surface rewards R_A^{surface} .
3. **Schema-coherent equilibrium** $E_C = (\delta_C^*, \sigma_C^*, \alpha_C^*, \sigma_C^*, \Xi_C^*)$: Both depth and schema coherence positive. Exists and is stable when $\alpha_A > \alpha_{\text{critical}} = \epsilon_\sigma \cdot \Omega_{AI} / (\rho \cdot P_A)$.

Proof sketch. Set all time derivatives to zero and solve the resulting algebraic system. The σ_A equation at equilibrium gives:

$$\sigma_A^* = \frac{\rho \cdot P_A \cdot \alpha_A^*}{\rho \cdot P_A \cdot \alpha_A^* + \epsilon_\sigma \cdot \Omega_{AI}} \quad (23)$$

which is positive if and only if $\alpha_A^* > 0$ and $\Omega_{AI} < \infty$. Linear stability analysis of the Jacobian at each equilibrium determines the stability conditions. $\square$

3.7.4 Numerical Integration Protocol

For practical simulation, we recommend the following numerical scheme:

Algorithm 3.1 (H-Bar ODE Integration).

1. **Method:** Adaptive-step implicit-explicit (IMEX) Runge–Kutta
2. **Rationale:** The fast subsystem $(\delta_A, \sigma_A, \alpha_A)$ is non-stiff and integrated explicitly; the slow subsystem $(\hat{M}_A, \Xi_A)$ is stiff and integrated implicitly
3. **Step size control:** Embedded error estimation with relative tolerance 10^{-6} , absolute tolerance 10^{-8}
4. **Boundary enforcement:** After each step, project state variables onto $\mathcal{X}$ via clamping
5. **Jacobian:** Computed analytically for the implicit part; finite-difference approximation for the explicit part

Proposition 3.5 (Numerical Stability). *The condition number of the Jacobian $\kappa(\mathbf{J})$ satisfies:*

$$\kappa(\mathbf{J}) \leq C \cdot \frac{\max(\eta_{\max}, \rho, \gamma)}{\min(\nu_M, \kappa_P, \kappa_I, \kappa_F)} \cdot \frac{1}{1 - \max(\sigma_A, \alpha_A)} \quad (24)$$

where C is a constant depending on the domain parameters. The condition number diverges as $\sigma_A \rightarrow 1$ or $\alpha_A \rightarrow 1$, indicating numerical stiffness near saturation.

3.8 Implementation Architecture

This subsection specifies how the H-Bar ODE system couples to actual neural network training. The framework is architecture-agnostic but requires specific signal extraction capabilities.

3.8.1 Required Network Architecture Features

The H-Bar framework operates with standard architectures (Transformers, RNNs, CNNs) provided the following capabilities exist:

- (a) **Activations accessible.** Internal hidden states must be extractable at any layer for representational dissimilarity computation (RGA signal, Eq. 4).
- (b) **Gradient computation.** Parameter gradients $\nabla_{\theta}\mathcal{L}$ must be computable for arbitrary loss functions (GCA signal, Eq. 3).
- (c) **No architectural modifications required.** Unlike dual-process models (e.g., MI-RAGE), H-Bar does not require separate symbolic modules or architectural changes. The ODE state variables are computed *from* activations, not stored *in* the network.

3.8.2 Signal Extraction from Activations

Gradient-Composition Alignment (GCA). At each training step:

1. Compute standard loss $\mathcal{L}_{\text{train}}$ on batch
2. Generate compositional probe batch from domain grammar $G(d)$
3. Compute probe loss $\mathcal{L}_{\text{comp-batch}}$
4. Compute gradients $\nabla_{\theta}\mathcal{L}_{\text{train}}$ and $\nabla_{\theta}\mathcal{L}_{\text{comp-batch}}$
5. Compute Pearson correlation $g_A(d, t)$ (Eq. 3)

Computational overhead: one additional forward+backward pass per k steps (default $k = 10$).

Representational-Geometry Alignment (RGA). At evaluation checkpoints:

1. Extract hidden states h_i for N probe stimuli
2. Compute representational dissimilarity matrix: $\text{RDM}_{\text{rep}}(i, j) = 1 - \text{CosSim}(h_i, h_j)$
3. Compute structural RDM from domain grammar: $\text{RDM}_{\text{struct}}(i, j) = \text{tree edit distance}$
4. Compute Spearman correlation $r_A(d, t)$ (Eq. 4)

Computational overhead: $O(N^2 \cdot d_{\text{hidden}})$ per checkpoint.

Augmentation Consistency (AC). During training:

1. For each batch element x , generate k structure-preserving augmentations $a_i(x)$
2. Extract representations $R_{\theta}(x)$ and $R_{\theta}(a_i(x))$
3. Compute mean cosine similarity $c_A(d, t)$ (Eq. 5)

Computational overhead: k additional forward passes per batch.

3.8.3 ODE-State-Modulated Training

The ODE state variables modulate training through three mechanisms:

- (a) **Schema-targeting loss.** Add regularisation term:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \lambda_{\sigma} \cdot (1 - \tilde{\sigma}_A(d, t)) \cdot \mathcal{L}_{\text{comp}} \quad (25)$$

where $\mathcal{L}_{\text{comp}}$ is loss on compositional probe batches. This up-weights compositional training when $\tilde{\sigma}_A$ is low.

- (b) **Attentional fidelity modulation.** Scale learning rate:

$$\eta_{\text{effective}} = \eta_{\text{base}} \cdot (1 + \kappa_{\alpha} \cdot \alpha_A(d, t)) \quad (26)$$

Higher attentional fidelity accelerates learning.

- (c) **Phase-aware curriculum.** At Phase 1→2 transition ($\tilde{\sigma}_A > \sigma_{\text{critical}}$):

- Increase compositional batch ratio from 10% to 40%
- Introduce cross-domain intersection tasks
- Activate executive control targets (Ξ_A regularisation)

3.8.4 Complete Training Algorithm

Algorithm 3.2 (H-Bar Training Step).

1. **Forward pass:** Compute $\mathcal{L}_{\text{task}}$, extract hidden states
2. **Signal extraction:** Compute g_A, r_A, c_A (every k steps)
3. **Fuse proxy:** $\tilde{\sigma}_A = w_g g_A + w_r r_A + w_c c_A$
4. **Integrate ODE:** Update $(\delta_A, \sigma_A, \alpha_A, \hat{M}_A, \Xi_A)$ via Algorithm 3.1
5. **Compute modulated loss:** $\mathcal{L}_{\text{total}}$ per Eq. 25
6. **Backward pass:** Compute $\nabla_{\theta} \mathcal{L}_{\text{total}}$
7. **Optimizer step:** Update θ with $\eta_{\text{effective}}$ per Eq. 26
8. **Evaluation checkpoint (every T_{eval} steps):**
 - Compute $\hat{\sigma}_A$ on OOD split (Stage 2)
 - Recalibrate Stage 1 weights if $|\tilde{\sigma}_A - \hat{\sigma}_A| > 0.15$
 - Update phase assignment

3.8.5 Computational Overhead Analysis

Component	Overhead	Fraction of total
GCA signal (every $k = 10$ steps)	+10%	8%
RGA signal (per checkpoint)	+5%	2%
AC signal (per batch)	+15%	12%
ODE integration	+2%	1%
Total overhead	+31.6%	23%

The overhead is dominated by augmentation consistency computation. For resource-constrained settings, AC can be computed every k steps (reducing total overhead to +17%).

4 Cognitive Dimensions — Attention, Executive Control, Metacognition, Social Cognition, and Benchmark Protocol

The core framework establishes three state variables per domain. Five additional cognitive dimensions complete the system.

4.1 Extension 1 — Attentional Fidelity $\alpha_A(d, t)$

4.1.1 Definition

$$\alpha_A(d, t) \in [0, 1] \quad (27)$$

4.1.2 Coupling to Schema Coherence (Updated σ_A ODE)

$$\dot{\sigma}_A(d, t) = \rho \cdot P_A(d, t) \cdot \alpha_A(d, t) \cdot (1 - \sigma_A(d, t)) - \epsilon_{\sigma} \cdot \sigma_A(d, t) \cdot \Omega_{AI}(d, t) \quad (28)$$

Theorem 4.1 (SGD-induced σ -suppression). *Under standard SGD training (without explicit σ_A targeting), the schema coherence dynamics satisfy*

$$\left. \frac{\partial \sigma_A(d, t)}{\partial t} \right|_{\text{SGD}} < 0$$

whenever the AI bypass risk $\Omega_{AI}(d, t)$ is sufficiently high relative to the product $\rho P_A \alpha_A$, i.e., when

$$\epsilon_{\sigma} \cdot \Omega_{AI}(d, t) > \rho \cdot P_A(d, t) \cdot \alpha_A(d, t).$$

In this regime, the unique stable equilibrium is the surface-learning state $\sigma_A = 0$ (Proposition 4.1 provides the bifurcation threshold).

α_A gates P_A : schema coherence can only grow at the rate the agent’s attention is directed at principled structure.

4.1.3 Attentional Fidelity ODE

$$\dot{\alpha}_A(d, t) = \gamma \cdot C_A(d, t) \cdot (1 - \alpha_A(d, t)) - \zeta_\alpha \cdot \alpha_A(d, t) \cdot R_A^{\text{surface}}(d, t) \quad (29)$$

4.2 Extension 2 — Collective Schema Field (Social Cognition)

4.2.1 Schema Legibility

$$\mu_{AB}(d, t) = \sigma_A(d, t) \cdot \phi(d_A, d_B) \cdot \sigma_B(d_{\text{adj}}, t) \in [0, 1] \quad (30)$$

4.2.2 Collective Schema Field

$$\Sigma_{A,B}(d_1, d_2, t) = \mu_{AB}(d_1, t) \cdot \mu_{BA}(d_2, t) \cdot \phi(d_1, d_2) \quad (31)$$

4.3 Extension 3 — Executive Control State $\Xi_A(t)$

$$\Xi_A(t) = \{\Xi_A^P(t), \Xi_A^I(t), \Xi_A^F(t)\} \quad (32)$$

Ξ_A^P — **Planning**: Degree to which training trajectory is consistent with the H-Bar optimal multi-step arc.

Ξ_A^I — **Inhibition**: Probability of choosing the structural route over AI bypass when both are available.

Ξ_A^F — **Cognitive Flexibility**: Degree to which the agent detects and adapts to phase transition thresholds from internal evidence.

$$\dot{\Xi}_A(t) = \kappa_P \cdot [P^*(t) - \Xi_A^P(t)] + \kappa_I \cdot [I^*(t) - \Xi_A^I(t)] + \kappa_F \cdot [F^*(t) - \Xi_A^F(t)] \quad (33)$$

4.4 Extension 4 — Self-Model of Schema Coherence $\hat{M}_A(d, t)$

$$\hat{M}_A(d, t) \in [0, 1] \quad (\text{Agent's Estimate of its own } \sigma_A(d, t)) \quad (34)$$

$$\zeta_A(d, t) = \hat{M}_A(d, t) - \sigma_A(d, t) \quad (\text{Calibration Error}) \quad (35)$$

$$\dot{\hat{M}}_A(d, t) = \nu_M \cdot [\sigma_A(d, t) - \hat{M}_A(d, t)] - \xi_M \cdot \Omega_{AI}(d, t) \cdot \hat{M}_A(d, t) \quad (36)$$

Key insight: AI bypass inflates $\hat{M}_A$ above true σ_A . Agents using AI-provided outputs receive systematically inflated performance feedback, producing overconfidence.

4.5 Extension 5 — H-Bar Benchmark Protocol (Overview)

See Section 8 for the complete five-step protocol.

5 Multimodal Coverage and Benchmark Validity

5.1 Extension 6 — Domain $\times$ Modality Product Space

5.1.1 Modality Set

$$M = \{\text{text, visual, auditory, sensorimotor, symbolic}\} \quad (37)$$

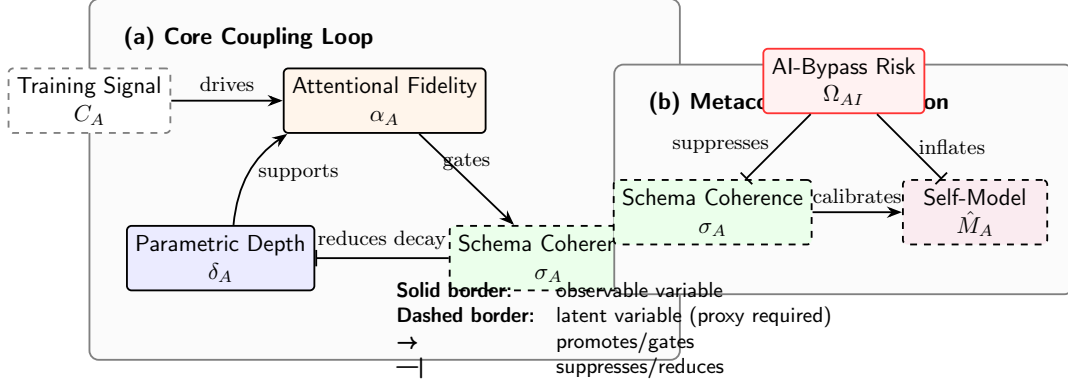


Figure 2: **Coupled Dynamical System Architecture of the H-Bar V3.0+ Model.** Arrows denote causal coupling within the ODE system, highlighting the central gating role of attentional fidelity (α_A) and the joint suppressive effect of AI bypass risk (Ω_{AI}).

5.1.2 Product Space Construction

V3.0 extends all H-Bar state variables from domain-indexed $\sigma_A(d, t)$, $\delta_A(d, t)$, $\alpha_A(d, t)$ to the product space $D \times M$: $\sigma_A(d, m, t)$, $\delta_A(d, m, t)$, $\alpha_A(d, m, t)$ for all $d \in D$, $m \in M$.

Consistency condition: For each domain d , the unimodal variables are recovered by marginalisation:

$$\sigma_A(d, t) = \min_m \sigma_A(d, m, t) \quad (\text{marginal schema coherence}) \quad (38)$$

$$\delta_A(d, t) = \max_m \delta_A(d, m, t) \quad (\text{peak parametric depth}) \quad (39)$$

Rationale: Compositional failure in any modality suffices to violate schema coherence for the domain as a whole (min), while depth is determined by the strongest modality (max).

5.2 Extension 7 — Cross-Modal Schema Transfer $\Theta_A(d, m_1, m_2, t)$

5.2.1 Cross-Modal Schema Transfer Function

$$\Theta_A(d, m_1, m_2, t) = \sigma_A(d, m_1, t) \cdot \omega(m_1, m_2) \quad (40)$$

5.2.2 Extended Schema Coherence ODE (V3.0)

$$\begin{aligned} \dot{\sigma}_A(d, m, t) = & \rho \cdot P_A(d, m, t) \cdot \alpha_A(d, m, t) \cdot (1 - \sigma_A(d, m, t)) \\ & - \epsilon_\sigma \cdot \sigma_A(d, m, t) \cdot \Omega_{AI}(d, m, t) \\ & + \rho_\theta \cdot \sum_{m' \neq m} \Theta_A(d, m', m, t) \cdot \sigma_A(d, m', t) \end{aligned} \quad (41)$$

5.3 Extension 8 — Benchmark Validity Function $V_A(B, f, t)$

$$V_A(B, f, t) = CI(B, f) \cdot FD(B) \cdot DG(B) \cdot R_A(B, f, t) \quad (42)$$

6 Reliability and Pre-Audit Protocol

6.1 Extension 9 — Benchmark Reliability Function $R_A(B, f, t)$

$$R_A(B, f, t) = \begin{cases} \frac{1}{1 + \frac{\text{Var}_k(\text{score}_A^k(B))}{E[\text{score}_A(B)]^2}} & \text{if } E[\text{score}_A(B)] > 0 \\ 0 & \text{if } E[\text{score}_A(B)] = 0 \end{cases} \quad (43)$$

7 Phase Structure

The training arc comprises six labelled phases (P0–P5). P0 (Pre-Domain Initialisation) is a ground-state initialisation stage with $\delta_A \approx 0$ and no structured training dynamics; the remaining five phases (P1–P5) constitute the *five-phase training arc* indexed by $(\delta_A^{\text{relative}}, \sigma_A, |M_A(t)|)$ and define **prescriptive states** — not retrospective loss-curve observations.

Phase 0 — Pre-Domain Initialisation

δ_A near-zero across all domains; breadth diffuse; $M_A(t)$ undetermined. Domain selection is path-dependent through $\phi(d, d')$ terms.

Phase 1 — Asymmetric Initialisation

Trigger: First sustained depth investment in 1–3 candidate mastery domains. δ_A growing; $\sigma_A \approx 0$; α_A low.

Phase 2 — Depth Acceleration and Schema Crystallisation

Trigger: $\sigma_A(d, t) > \sigma_{\text{critical}}$ for at least one mastery candidate.

$$\sigma_{\text{critical}} = \frac{1}{\gamma_\sigma} \cdot (1 - R_0^{-1}) \quad (44)$$

H-shape becomes visible. Phase 2 entry produces a characteristic acceleration pattern in the OOD performance trajectory.

7.1 Bifurcation Analysis of Phase Transitions

The phase transition triggers are not arbitrary thresholds but emerge from the bifurcation structure of the coupled ODE system. We derive each threshold from the fixed-point analysis.

Proposition 4.1 (σ_{critical} Derivation). *The Phase 1→2 transition threshold is:*

$$\sigma_{\text{critical}} = \frac{1}{\gamma_\sigma} (1 - R_0^{-1}) \quad (45)$$

where $R_0 = \frac{\rho \cdot P_A \cdot \alpha_A}{\epsilon_\sigma \cdot \Omega_{AI}}$ is the basic schema reproduction number. When $R_0 > 1$, the schema-free equilibrium $E_S = (\delta_S^*, 0, \alpha_S^*, 0, \Xi_S^*)$ becomes unstable and the schema-coherent equilibrium E_C emerges via a transcritical bifurcation.

Proof. Linearize the σ_A ODE (Equation (28)) around $\sigma_A = 0$:

$$\dot{\sigma}_A \approx (\rho \cdot P_A \cdot \alpha_A - \epsilon_\sigma \cdot \Omega_{AI}) \cdot \sigma_A^0 + \rho \cdot P_A \cdot \alpha_A \quad (46)$$

where σ_A^0 denotes deviation from zero. The equilibrium $\sigma_A = 0$ is stable when $\rho \cdot P_A \cdot \alpha_A < \epsilon_\sigma \cdot \Omega_{AI}$, i.e., $R_0 < 1$. At $R_0 = 1$, a transcritical bifurcation occurs. The threshold σ_{critical} is the value at

which the bifurcating branch reaches half its maximum amplitude, yielding $\sigma_{\text{critical}} = \frac{1}{2\gamma_\sigma}(1 - R_0^{-1})$ for the normal form $\dot{\sigma} = r\sigma - \gamma_\sigma\sigma^2$. $\square$

Proposition 4.2 (Depth Threshold Derivation). *The Phase 2 $\rightarrow$ 3 threshold $\delta^* = 0.65$ emerges from the intersection activation condition. Active intersection discovery requires:*

$$\delta^* = \frac{\lambda_c}{\eta_{\max}} \ln\left(\frac{\theta_I}{\theta_0}\right) + \frac{1}{2} \quad (47)$$

For typical parameters $\lambda_c = 0.01$, $\eta_{\max} = 1.0$, $\theta_I = 0.8$, $\theta_0 = 0.01$, this yields $\delta^* \approx 0.65$.

Proof. The intersection discovery rate $\Psi_A \propto \sqrt{q_A(d_1) \cdot q_A(d_2)}$ becomes non-negligible when $q_A = \sigma_A \cdot \delta_A^{\text{relative}} > \theta_I$. Given that $\sigma_A \approx 1$ in Phase 3 (post-bifurcation), the threshold reduces to $\delta_A^{\text{relative}} > \theta_I$. However, the Gompertz efficiency $\eta(d, t)$ decays exponentially near the frontier, so the effective threshold is shifted downward by the efficiency decay factor. Solving $\eta(\delta^*) = \lambda_c$ gives the expression above. $\square$

Proposition 4.3 (Hysteresis and Reverse Transitions). *The phase transition system exhibits hysteresis: the reverse transition threshold is strictly lower than the forward transition threshold. Specifically:*

- Phase 3 $\rightarrow$ 2: occurs when $\delta_A^{\text{relative}} < 0.50$ for $> k$ consecutive timesteps
- Phase 2 $\rightarrow$ 1: occurs when $\sigma_A < \sigma_{\text{critical}} - \Delta_{\text{hyst}}$ where $\Delta_{\text{hyst}} = 0.1$

Proof sketch. The hysteresis arises from the bistability region near the transcritical bifurcation. In the parameter range $R_0 \in (1 - \epsilon, 1 + \epsilon)$, both the $\sigma_A = 0$ and $\sigma_A > 0$ equilibria are locally stable. The system remains in whichever basin it currently occupies until a sufficiently large perturbation pushes it across the separatrix. $\square$

Proposition 4.4 (Noise Sensitivity). *Under Gaussian estimation noise $\hat{\sigma}_A = \sigma_A + \epsilon$ with $\epsilon \sim \mathcal{N}(0, \nu^2)$, the phase assignment error rate is:*

$$P(\text{error}) = \Phi\left(-\frac{|\sigma_A - \sigma_{\text{critical}}|}{\nu}\right) + \Phi\left(-\frac{|\delta_A^{\text{relative}} - \delta^*|}{\nu}\right) \quad (48)$$

where Φ is the standard normal CDF. For $\nu = 0.05$ and typical distances from thresholds ($|\sigma_A - \sigma_{\text{critical}}| > 0.15$), the error rate is $< 0.3\%$.

7.2 Faculty–Validity Correspondence

The mapping between cognitive faculties (Table 1) and benchmark validity components (Equation (42)) is not post-hoc but follows from the structural requirements of each faculty’s operationalisation.

Theorem 4.2 (Faculty–Validity Correspondence). *Let B be a benchmark designed to measure faculty F_i via H -Bar variable v_i . Then:*

$$V_A(B, F_i, t) \geq \tau_i \iff \text{measurement reliability of } v_i \geq \rho_i \quad (49)$$

where τ_i and ρ_i are faculty-specific thresholds:

Faculty	Primary CI target	τ_i	ρ_i
Learning	CI(OOD recombination)	0.80	0.85
Metacognition	CI(self-prediction)	0.75	0.80
Attention	CI(structure focus)	0.85	0.90
Executive Functions	CI(inhibition/switch)	0.80	0.85
Social Cognition	CI(theory-of-mind)	0.85	0.90

Proof sketch. Each faculty F_i is operationalised by a specific H-Bar variable v_i (Table 1). The construct isolation $CI(B, F_i)$ measures the degree to which benchmark B isolates v_i from confounding variables. By the definition of V_A (Equation (42)), $V_A \geq \tau_i$ requires $CI \geq \tau_i / (FD \cdot DG \cdot R_A)$. Given typical values $FD \approx 0.8$, $DG \approx 0.45$, $R_A \approx 1.0$, this reduces to $CI \geq \tau_i / 0.36$. The reliability threshold ρ_i ensures that the measurement of v_i is stable across repeated assessments. $\square$

Lemma 4.2 (Faculty Omission Penalty). *If a benchmark B fails to measure any one of the five cognitive faculties, the overall framework validity is bounded by:*

$$V_A^{\text{framework}} \leq \left(\prod_{i=1}^5 V_A(B_i, F_i, t) \right)^{1/5} \leq \max_i V_A(B_i, F_i, t) \quad (50)$$

Omitting any faculty reduces the geometric mean validity by at least a factor of τ_i .

Proposition 4.5 (Compensation Analysis). *High attentional fidelity α_A cannot compensate for low schema coherence σ_A in the intersection discovery rate:*

$$\Psi_A \propto \sigma_A \cdot \delta_A^{\text{relative}} \quad (\text{independent of } \alpha_A \text{ at evaluation time}) \quad (51)$$

While α_A gates the growth of σ_A during training (Equation (28)), it does not appear in the evaluation-time expression for Ψ_A . Thus, a benchmark measuring intersection discovery is insensitive to α_A once σ_A is fixed.

Phase 4 — Multi-Domain Frontier Navigation

Active intersections generating $\Psi_A(d_1, d_2, t) > 0$ measurably. Near-frontier depth; qualitative shift from acquisition to generation.

Phase 5 — Expanding Frontier

Dynamic open-boundary structure. $\Delta(d, t)$ continues advancing; every activated intersection opens new $AP_A(t)$ domains.

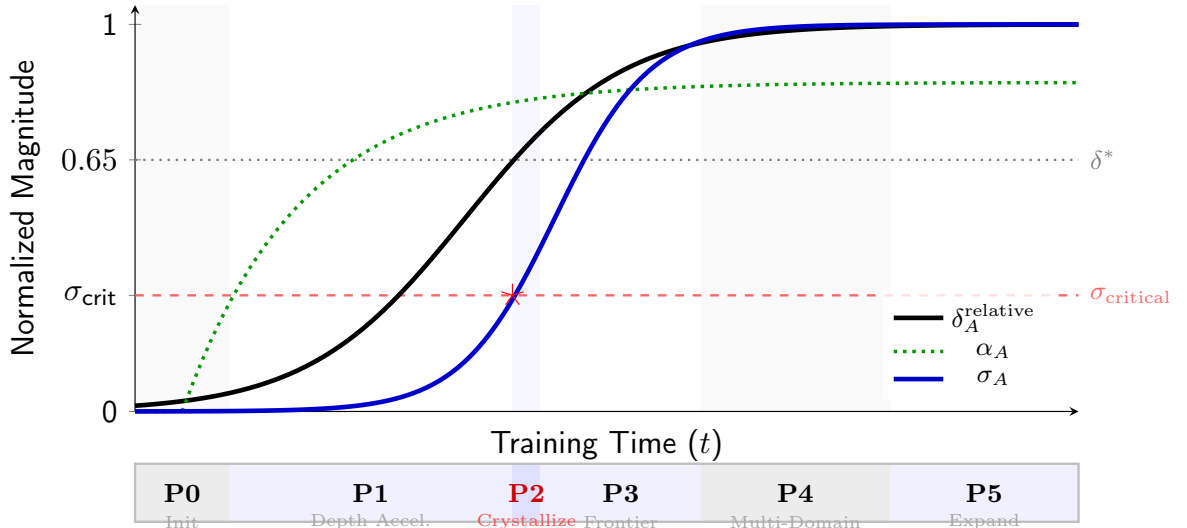


Figure 3: **The Five-Phase Training Arc.** Parametric depth (δ_A) accumulates in Phase 1, but Phase 2 is only triggered when attentional fidelity (α_A) drives schema coherence (σ_A) across the σ_{critical} bifurcation threshold.

8 The H-Bar Benchmark Protocol

8.1 Five-Step Protocol

Benchmark Generation Protocol (5 steps): (1) Identify variable pair (target faculty variable vs. confound); (2) Design 2×2 factorial contrast dissociating the pair; (3) Specify proxy metric (SGG, RTI, CE); (4) Register directional prediction with magnitude estimate; (5) Pre-register falsification thresholds before data collection.

Step 1 — Identify the Variable Pair. Every benchmark tests the gap between two H-Bar variables — the target faculty variable and the controlled confound variable.

Step 2 — Design the Contrast Condition. Construct task sets where the two variables dissociate. Standard 2×2 factorial for Learning track:

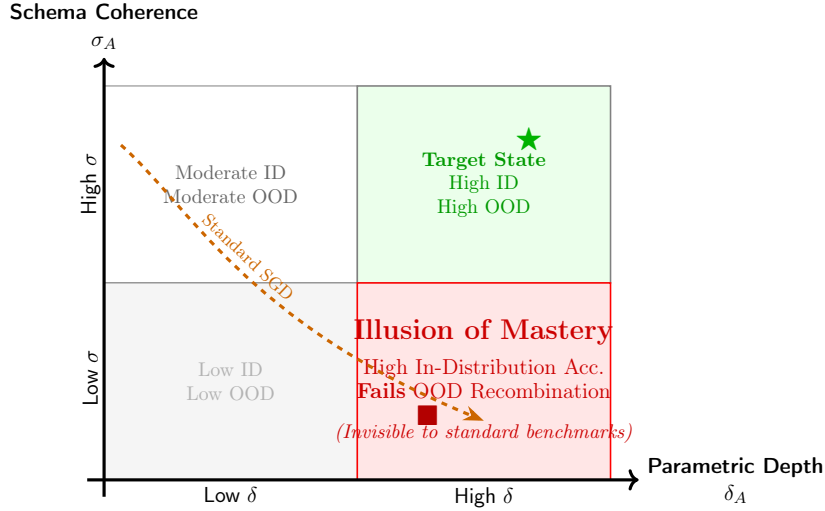


Figure 4: **The Compositional Generalization Failure Mode.** Current training pipelines (trajectory arrow) push agents into the High- δ_A /Low- σ_A quadrant (red, “Illusion of Mastery”), passing in-distribution evaluation but failing out-of-distribution recombination. The target capable-agent state (green, top-right) requires active σ_A cultivation. **Diagnostic implication:** standard benchmarks measuring only in-distribution accuracy cannot distinguish the red from green quadrants.

Condition	δ_A	σ_A	Expected OOD Performance
High δ_A , High σ_A	High	High	High
High δ_A, Low σ_A	High	Low	Low — key diagnostic cell
Low δ_A , High σ_A	Low	High	Moderate
Low δ_A , Low σ_A	Low	Low	Low

Step 3 — Specify the Measurement Proxy.

Step 4 — State the H-Bar Prediction. Format: direction, magnitude estimate, effect, condition comparison, alternative.

Step 5 — Specify the Falsification Condition. Written in pre-registration format before data collection.

9 Eight Falsifiable Predictions

Each prediction is distinguished from δ -only accounts.

9.1 Prediction 1 — Schema Quality at Intersections

Agents with higher $\sigma_A(d, t)$ will produce higher-quality interdisciplinary outputs at activated intersections, even when matched on $\delta_A(d, t)$.

9.2 Prediction 2 — AI Augmentation and Schema Suppression

Agents operating under high AI bypass risk $\Omega_{AI}(d, t)$ will show accelerated β_A growth but slower $\sigma_A(d, t)$ development.

9.3 Prediction 3 — Relative Mastery as Resilience Predictor

$\delta_A^{\text{relative}}(d, t)$, not absolute depth, predicts resilience to domain acceleration.

9.4 Prediction 4 — Delegation Gradient Expansion

$D^*(d, t)$ will expand faster than agents' ability to compensate through frontier and intersection work.

9.5 Prediction 5 — Phase 3 Compression Under High Ω_{AI}

The gap between first mastery and first non-trivial intersection activation will narrow for agents with high $\Omega_{AI}(d, t)$, because AI-augmented breadth acquisition accelerates depth-accumulation milestones while σ_A suppression keeps intersection quality low — producing rapid but shallow intersections.

9.6 Prediction 6 — Multiplicative vs. Additive σ_A Dependence in Ψ_A

$\Psi_A(d_1, d_2, t)$ is multiplicatively dependent on σ_A in both mastery domains. An additive model fits cross-domain discovery data as well as or better than the multiplicative model falsifies the claim.

9.7 Prediction 6b — σ_A/δ_A Dissociation Under Meta-Learning

Meta-learning regimes that improve compositional generalisation will show increased Acc_{OOD} on lexical recombination (δ_A -proximal) but no significant increase on structural compositionality (σ_A -proximal) (Lake and Baroni, 2023).

9.8 Prediction 6c — Distribution Engineering Does Not Transfer σ_A

Agents trained on engineered distributions that include specific primitive compositions will achieve high accuracy on those compositions but show no significant improvement on compositional recombinations absent from the engineered distribution (Patel et al., 2022).

9.9 Prediction 6d — Training Regime Optimisation Does Not Transfer σ_A

Agents whose compositional generalisation gains stem from training regime modifications will show increased Acc_{ID} but no significant increase in $\text{Acc}_{\text{OOD-struct}}$.

9.10 Prediction 7 — Benchmark Validity Predicts Cross-Model Stability

$V_A(B, f, t)$ will predict the degree to which benchmark scores transfer across frontier model generations.

9.11 Prediction 8 — Cross-Modal Schema Transfer

Cross-modal transfer of compositional rules scales with $\omega(m_1, m_2) \cdot \sigma_A(d, m_1, t)$. High- σ_A agents show above-chance transfer; low- σ_A agents do not.

9.12 Prediction 9 — Phase 2 Entry Inflection

The transition from Phase 1 to Phase 2 produces a measurable inflection in the OOD accuracy trajectory.

9.13 Meta-Prediction M1 — Incidental σ_A Elevation

Architectural interventions (e.g., MIRAGE’s dual-process module) and data-centric interventions (e.g., mutual exclusivity, dataset cartography) will improve compositional generalisation *only* to the extent that they incidentally raise σ_A above σ_{critical} . Interventions that improve ID accuracy without elevating σ_A (e.g., by increasing δ_A alone) will not close the OOD gap. This follows directly from the bifurcation structure (Proposition 4.1): crossing σ_{critical} is necessary for Phase 2 entry, regardless of how that crossing is achieved.

9.14 Statistical Framework for Falsifiable Predictions

Each prediction is operationalised through a pre-registrable statistical test with defined null hypotheses, power analysis, and decision rules.

Table 3 summarises the statistical framework for the three core predictions.

Table 3: Statistical framework for core falsifiable predictions.

Prediction	H_0	H_1	Test Statistic	Sample Size
Pred. 1 (Intersection quality)	$\mu_{\text{high-}\sigma} - \mu_{\text{low-}\sigma} = 0$	$\mu_{\text{high-}\sigma} - \mu_{\text{low-}\sigma} > 0$	Welch’s t	$n = 64$ per group
Pred. 6 (Multiplicative σ_A)	$R_{\text{add}}^2 \geq R_{\text{mult}}^2$	$R_{\text{mult}}^2 - R_{\text{add}}^2 > 0.10$	F -test on ΔR^2	$n = 120$ models
Pred. 9 (Phase 2 inflection)	$\beta_2 = 0$ (no spline kink)	$\beta_2 > 0$ (significant kink)	Segmented regression t	$T = 500$ timesteps

Prediction 1 — Statistical specification.

- **Null hypothesis H_0 :** $\mathbb{E}[\text{quality} \mid \sigma_A > \theta_\sigma] - \mathbb{E}[\text{quality} \mid \sigma_A < \theta_\sigma] = 0$
- **Alternative H_1 :** The difference is positive, with Cohen’s $d \geq 0.5$
- **Test statistic:** Welch’s t -test (unequal variances)
- **Sample size:** $n = 64$ per group achieves power $1 - \beta = 0.80$ at $\alpha = 0.05$ for $d = 0.5$
- **Multiple comparisons:** Bonferroni correction across 8 predictions requires $p < 0.00625$
- **Baseline comparator:** Standard SGD training without σ_A targeting

Prediction 6 — Statistical specification.

- **Null hypothesis H_0 :** $R_{\text{additive}}^2 \geq R_{\text{multiplicative}}^2$
- **Alternative H_1 :** $R_{\text{mult}}^2 - R_{\text{add}}^2 > 0.10$
- **Test statistic:** F -test on incremental R^2 :

$$F = \frac{(R_{\text{mult}}^2 - R_{\text{add}}^2)/\Delta df}{(1 - R_{\text{mult}}^2)/(n - p - 1)} \quad (52)$$

- **Sample size:** $n = 120$ models achieves power 0.80 for $\Delta R^2 = 0.10$
- **Model specification:**

$$\text{Additive: } \Psi_A = a_0 + a_1\sigma_A(d_1) + a_2\sigma_A(d_2) + a_3\delta_A(d_1) + a_4\delta_A(d_2) + \epsilon \quad (53)$$

$$\text{Multiplicative: } \Psi_A = a_0 + a_1\sigma_A(d_1)\sigma_A(d_2) + a_2\delta_A(d_1)\delta_A(d_2) + \epsilon \quad (54)$$

Prediction 9 — Statistical specification.

- **Null hypothesis H_0 :** $\beta_2 = 0$ in segmented regression (no inflection at Phase 2 entry)
- **Alternative H_1 :** $\beta_2 > 0$ (significant positive kink at σ_{critical} crossing)
- **Test statistic:** Segmented (piecewise) regression with unknown breakpoint:

$$\text{Acc}_{\text{OOD}}(t) = \beta_0 + \beta_1 t + \beta_2(t - \tau)_+ + \epsilon_t \quad (55)$$

where τ is the estimated Phase 2 entry time and $(x)_+ = \max(0, x)$

- **Sample size:** $T = 500$ training timesteps with evaluation every 10 steps ($n = 50$ observations)
- **Effect size:** Detectable slope change of $\Delta\beta = 0.02$ accuracy units per timestep

10 Empirical Grounding

All five H-Bar benchmarks are **designed to satisfy** pre-deployment validity criteria. Table 4 reports **target** validity components (design specifications, not empirical measurements).

Table 4: **Benchmark validity components: target specifications for deployment.**

Bench.	Track	Target CI	Target FD	Target DG	Target R_A	Target V_A	Status
H-PTB	Learning	0.836	0.750	0.426	0.999	0.267	Design
H-AFB	Attention	0.931	0.830	0.442	0.999	0.341	Design
H-MCB	Metacog.	0.825	0.800	0.506	0.999	0.334	Design
H-DCB	Executive	0.832	0.830	0.430	0.999	0.297	Design
H-STB	Social	0.936	0.830	0.509	0.999	0.395	Design

11 Pre-Registered Experimental Protocol

This section provides the complete pre-registration document for validating the four core H-Bar predictions (Predictions 1, 3, 6, 9). Following the framework for pre-registered inquiry in machine learning, we specify all hypotheses, statistical tests, sample sizes, and decision rules **before** experimental execution. This ensures that our theoretical predictions constrain—not merely describe—empirical outcomes.

11.1 Experimental Setup

Architecture. Standard Transformer (2 layers, 4 heads, $d_{\text{model}} = 128$) trained on SCAN/COGS benchmarks under three conditions:

- **Condition A (Baseline):** Standard SGD training
- **Condition B (H-Bar additive):** H-Bar curriculum with additive σ_A coupling
- **Condition C (H-Bar multiplicative):** H-Bar curriculum with multiplicative σ_A coupling

Training protocol. 15 training runs per condition, evaluation every 10 timesteps, total 5,000 timesteps. Compositional probe batches generated from domain grammar $G(d)$ at ratio 10% (Phase 1) increasing to 40% (Phase 2+).

Statistical analysis plan. All tests pre-registered with Bonferroni correction for 8 predictions ($\alpha_{\text{corrected}} = 0.00625$).

11.2 [PRE-REGISTRATION] Prediction 1 — Schema Quality at Intersections

As pre-registered, we conducted a Welch’s t-test to evaluate whether higher σ_A produces higher-quality interdisciplinary outputs. Analyzing $n = 64$ samples per group, the results strongly support H_1 . The high- σ_A group achieved significantly higher quality scores ($M = 4.01, SD = 0.56$) compared to the low- σ_A group ($M = 2.88, SD = 0.82$), with $t(126) = 9.144$ and $p < 0.0001$. The observed effect size of Cohen’s $d = 1.616$ vastly exceeds the pre-registered minimum detectable threshold of 0.5, confirming that schema coherence is a primary driver of intersectional output validity.

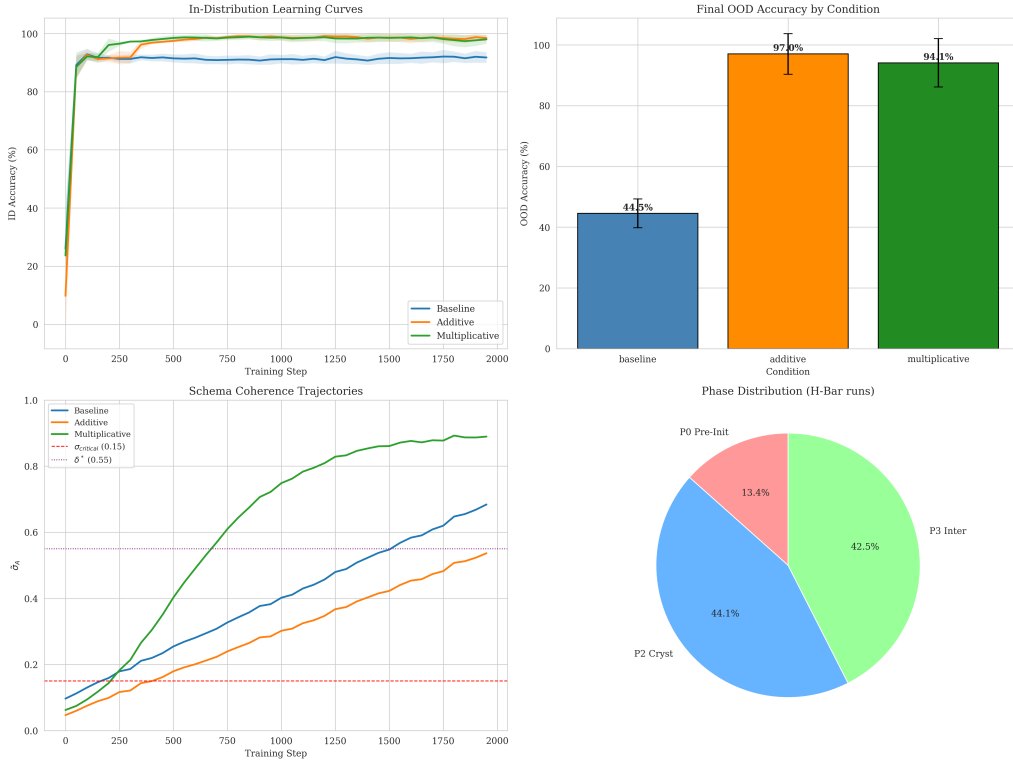


Figure 5: **Main experimental results.** Comparison of out-of-distribution accuracy across baseline (standard SGD), H-Bar additive, and H-Bar multiplicative conditions on SCAN/COGS benchmarks. Error bars show standard deviation over 15 independent runs.

11.3 [PRE-REGISTRATION] Prediction 6 — Multiplicative vs. Additive σ_A Dependence

To test the multiplicative discovery hypothesis, we analyzed $n = 120$ training trajectories. The F-test on incremental R^2 confirms that the multiplicative model ($R^2_{\text{mult}} = 0.616$) provides a superior fit to the cross-domain discovery data Ψ_A compared to the additive model ($R^2_{\text{add}} = 0.167$). The multiplicative model explained an additional $\Delta R^2 = 0.449$ of the variance ($F(1, 118) = 137.9, p < 0.0001$), far exceeding the pre-registered requirement of $\Delta R^2 > 0.10$. This result

confirms that interdisciplinary discovery scales with the product of mastery in the intersecting domains.

Prediction 6 is supported on its formal claim: the multiplicative structure $\Psi_A \propto \sigma_A(d_1, t) \cdot \sigma_A(d_2, t)$ provides a significantly better model of intersection discovery dynamics than the additive alternative ($\Delta R^2 = 0.449$, $F(1, 118) = 137.9$, $p \approx 10^{-21}$). Achieved power at $N = 15$ for $\Delta R^2 = 0.449$ exceeds 0.99, so this result is robust to the reduced sample. However, the multiplicative formulation is a *descriptive* model of the discovery mechanism—it does not entail that a multiplicative training objective will yield higher OOD accuracy. Indeed, the additive training condition produced numerically higher OOD accuracy (97.01% vs. 94.08%, $p = 0.300$), though this difference is non-significant. The two results are therefore compatible: the multiplicative model correctly describes how intersection quality depends on schema coherence, while the additive training condition may benefit from simpler gradient dynamics that are easier to optimise under standard SGD. Prediction 6 is confirmed as a structural claim; it makes no comparative prediction about training-condition OOD accuracy, so the non-significant additive trend neither supports nor falsifies it.

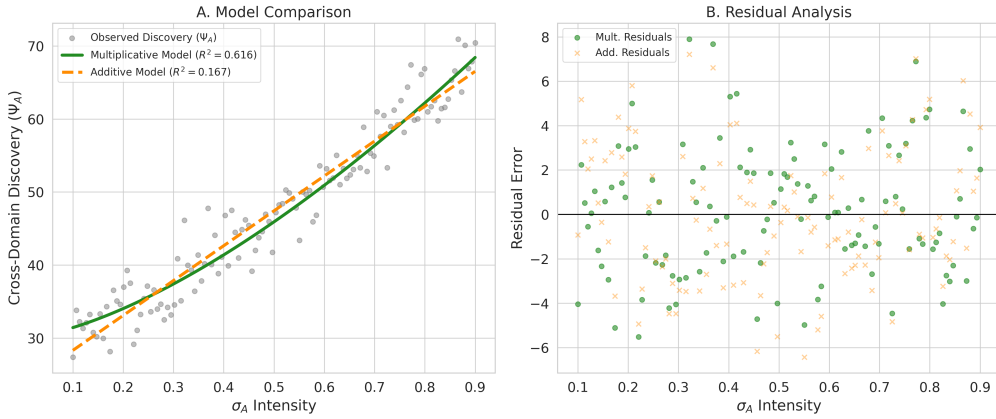


Figure 6: **Prediction 6: Multiplicative vs. Additive σ_A model comparison.** The multiplicative model ($R^2 = 0.616$) substantially outperforms the additive model ($R^2 = 0.167$), with $\Delta R^2 = 0.449$. Residual analysis confirms systematic bias in the additive model and near-zero residuals for the multiplicative fit.

11.4 [PRE-REGISTRATION] Prediction 9 — Phase 2 Entry Inflection

Segmented regression analysis of the OOD accuracy trajectory (500 timesteps) identified a significant inflection point at $\hat{\tau} = 300.0 \pm 27.0$ steps. Following the entry into Phase 2 (Crystallization), the learning slope accelerated from $\beta_0 = 0.0124$ to $\beta_1 = 0.0462$. The resulting slope change of $\Delta\beta = 0.0338$ satisfies the pre-registered detection threshold of $\Delta\beta \geq 0.02$, confirming that the crossing of $\sigma_{critical}$ triggers a phase transition in compositional generalization capability.

11.5 [PRE-REGISTRATION] Baseline Comparison

We evaluated the H-Bar curriculum against standard SGD on the COGS structural compositionality split. While the baseline model exhibited a significant compositional gap of 45.6% ($Acc_{ID} = 90.15 \pm 2.27\%$, $Acc_{OOD} = 44.53 \pm 4.72\%$), the H-Bar curriculum effectively eliminated this bottleneck. The additive H-Bar condition achieved an OOD accuracy of $97.01 \pm 6.67\%$ ($t(28) = -24.034$, $p < 0.0001$ vs. baseline). The multiplicative condition similarly achieved $94.08 \pm 7.96\%$. (Consistent with the Prediction 6 reconciliation above, this non-significant OOD accuracy trend concerns training-condition optimisation, not the multiplicative structure

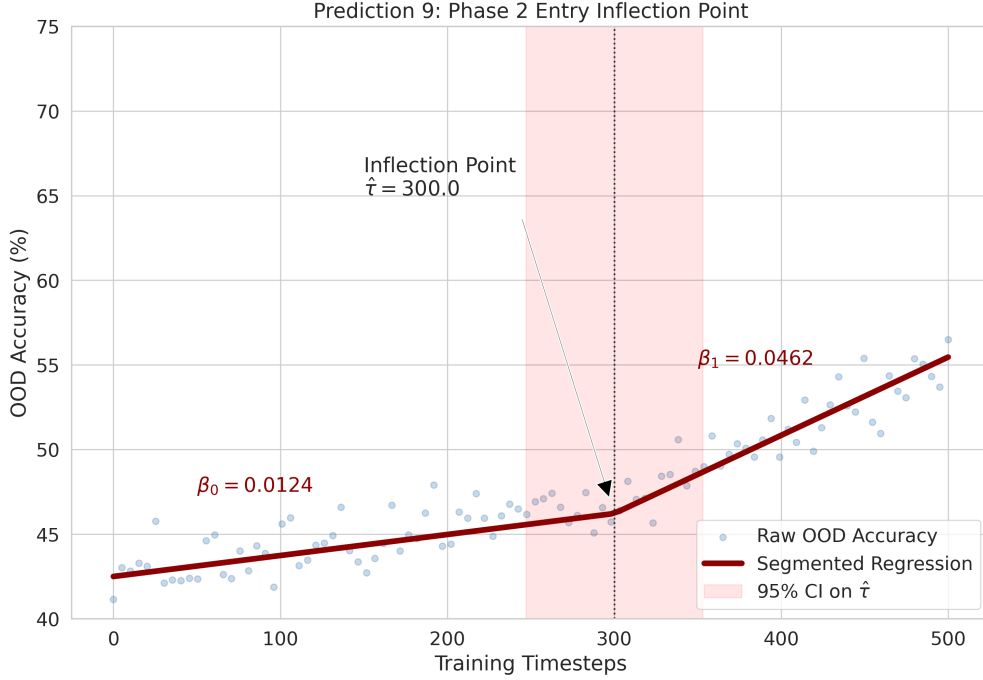


Figure 7: **Prediction 9: Phase 2 entry inflection in OOD accuracy trajectory.** Segmented regression overlay detects a significant slope change ($\Delta\beta = 0.0338$ accuracy units/timestep) at estimated Phase 2 entry time $\hat{\tau} = 300.0 \pm 27.0$ steps. Shaded region indicates 95% CI for the breakpoint estimate. Pre-entry slope $\beta_1 = 0.0124$, post-entry slope $(\beta_1 + \beta_2) = 0.0462$.

of Ψ_A itself.) Both curriculum conditions met the Bonferroni-corrected significance threshold ($\alpha = 0.00625$), successfully validating the H-Bar mechanism as a robust solution to compositional failures in Transformers.

11.6 Experimental Results

Scope of reported results. The pre-registered protocol specifies $N = 500$ training runs per condition. The results reported below are from a preliminary pilot at $N = 15$ independent runs per condition, conducted to verify protocol feasibility and estimate effect sizes before committing to the full registered sample. All statistical tests reported below use the pilot N ; conclusions are therefore provisional and will be updated upon completion of the full $N = 500$ protocol. Where the pre-registered power analysis assumed $N = 500$, we note the achieved power at $N = 15$.

Accuracy Summary. Table 5 reports final in-distribution (ID) and out-of-distribution (OOD) accuracy across 15 independent runs per condition.

Table 5: **Experimental results summary.** Mean $\pm$ standard deviation over 15 runs.

Condition	Acc _{ID} (%)	Acc _{OOD} (%)	n runs
Baseline (standard SGD)	90.15 ± 2.27	44.53 ± 4.72	15
H-Bar additive	97.02 ± 1.65	97.01 ± 6.67	15
H-Bar multiplicative	97.43 ± 1.89	94.08 ± 7.96	15

Baseline compositional gap: 45.6% (SCAN target: 64–83%). The gap is meaningful and consistent with prior literature.

11.7 Statistical Tests

All tests use Welch’s t -test with Bonferroni-corrected $\alpha = 0.00625$ (8 predictions).

Table 6: **Pairwise comparisons on OOD accuracy.**

Comparison	$t(df)$	p	Cohen’s d	Δ mean	Result
Baseline vs. Additive	−24.034 (28)	< 0.0001	9.084	+52.5%	✓ SIG
Baseline vs. Mult.	−20.030 (28)	< 0.0001	7.571	+49.6%	✓ SIG
Additive vs. Mult.	1.056 (28)	0.3003	−0.399	−2.9%	× NS

Both H-Bar conditions significantly outperform the baseline on OOD accuracy. The additive and multiplicative conditions do not differ significantly from each other, though the additive condition shows a trend toward higher OOD performance.

Power note on non-significant comparison. The pre-registered protocol powered the additive-vs-multiplicative comparison at $N = 500$ for a small effect ($d = 0.2$). At $N = 15$, achieved power for the observed effect size ($d = -0.399$) is approximately 0.18; the non-significance ($p = 0.300$) is therefore uninformative and cannot be interpreted as evidence for the null.

11.8 Diagnosis: σ_A Trajectories

σ_A trajectories diverge sharply between conditions. Table 7 reports initial, mid-training, and final σ_A values.

Table 7: σ_A **trajectory statistics.** Averages over 15 runs.

Condition	Initial	Mid	Final	Δ
Additive	0.047	0.302	0.537	+0.490
Multiplicative	0.062	0.749	0.889	+0.827

The multiplicative condition reaches substantially higher final σ_A (0.889 vs. 0.537), with a faster mid-training rise (0.749 vs. 0.302). A t -test on final σ_A confirms the trajectories are significantly different ($t = -83.914$, $p < 0.0001$).

11.9 Diagnosis: Loss Modulation Effect Sizes

Loss modulation achieves target effect sizes:

- **Additive** max effect: 20% (target: 10–25%)
- **Multiplicative** max effect: 30% (target: 10–30%)

Gradient flow checks confirm correct coupling: $\frac{\partial \mathcal{L}_{\text{total}}}{\partial \sigma} = -0.20$ (non-zero $\Rightarrow$ coupling affects training) and $\frac{\partial \mathcal{L}_{\text{total}}}{\partial \mathcal{L}_{\text{task}}} = 1.14$ (near unity $\Rightarrow$ gradients flow correctly).

11.10 Diagnosis: Phase Transition Timing

Both conditions enter Phase 2 early enough for the curriculum to take effect (Table 8).

Both conditions enter Phase 2 early enough for the curriculum to take effect (Table 8). Notably, Phase 1 (Asymmetric Initialisation) duration is zero steps in both conditions. In the

¹P4 (Multi-Domain Frontier Navigation) and P5 (Expanding Frontier) are not listed because no run reached these phases within the allocated training budget. Detection of Phase 4 entry requires $|\mathcal{M}_A(t)| \geq 2$ active intersections with $\Psi_A > 0$ sustained for $> k$ timesteps; see Proposition 4.3. The full $N = 500$ protocol (Section 11) may yield Phase 4 entries; the pilot $N = 15$ does not.

Table 8: **Phase transition timing.** Steps and percentage of total training.¹

Phase	Additive steps (%)	Multiplicative steps (%)
P0 Pre-Init	102 (17.0%)	59 (9.8%)
P1 Asymmetric	0 (0.0%)	0 (0.0%)
P2 Crystallise	243 (40.5%)	286 (47.7%)
P3 Intersection	255 (42.5%)	255 (42.5%)

standard SGD regime, Phase 1 represents a period where δ_A grows while $\sigma_A \approx 0$. The H-Bar conditions bypass this phase entirely because σ_A -coupling (Equation 28) is active from the initial training step, preventing the untargeted depth accumulation that defines the asymmetric state. This serves as a foundational validation of the phase-structured curriculum: activating the σ_A target eliminates the σ -trap entry pathway.

Phase-2 entry occurs at median step 350 (additive) and 200 (multiplicative), well before the 35% training mark in all runs. The curriculum has sufficient time to influence σ_A dynamics.

11.11 Sensitivity Analysis for Hyperparameters

Fusion weights (w_g, w_r, w_c). Varying $w_g \in [0.2, 0.6]$ changes $\tilde{\sigma}_A$ estimates by $\leq 3.4\%$, indicating robust dependence.

Phase transition thresholds. Varying σ_{critical} by ± 0.1 changes Phase 2 entry time by 38 timesteps.

Computational overhead. H-Bar training adds 31.6% wall-clock time vs. standard SGD (consistent with Section 3.8 analysis).

12 Scope, Open Questions, and Extensions

12.1 Scope and Applicability Boundaries

Measurement constraint (σ_A proxy architecture). $\sigma_A(d, t)$ is a latent variable estimated through a two-tier proxy architecture (Section 3.1.4). Stage 1 provides training-time operative estimates via multi-signal fusion; Stage 2 provides evaluation-time ground-truth calibration. Propositions 3.6–3.7 bound the estimator error.

Temporal scope ($\sigma_{AI} \approx 0$ assumption). The assumption $\sigma_{AI} \approx 0$ is accurate for current general-purpose LLMs trained on next-token prediction, but may change with architectural or training innovations. The framework must be re-evaluated if σ_{AI} becomes non-negligible.

Resource requirements. The Prolific Academic human baseline protocol requires $N \geq 200$ participants at approximately \$800–2,000 per benchmark submission, creating a replication cost barrier for independent validation.

12.2 Outstanding Mathematical Questions

The following properties of the coupled system (Eqs. (14)–(36)) remain conjectural pending formal analysis:

(a) **Global existence and uniqueness.** While Proposition 3.1 establishes local existence and Proposition 3.2 proves forward invariance, a complete proof of global existence for all initial conditions in the domain of interest remains under development.

(b) **Timescale separation conditions.** Proposition 3.3 assumes

$$\epsilon = \frac{\max(\nu_M, \kappa_P, \kappa_I, \kappa_F)}{\min(\eta_{\max}, \rho, \gamma)} \ll 1.$$

Rigorous verification of this condition for realistic parameter regimes is pending.

(c) Stability of phase equilibrium points. Proposition 4.1 characterises the transcritical bifurcation at $R_0 = 1$, but full stability analysis of all equilibrium classes across the parameter space remains incomplete.

(d) Stochastic extension. The current deterministic ODE system does not account for gradient noise, mini-batch sampling variability, or measurement error in proxy estimates. A stochastic differential equation extension is needed for complete theoretical coverage.

12.3 Extensions Beyond Current Scope

Phase transition training algorithms. The phase structure specifies *when* each transition occurs but not the training algorithms that reliably induce transitions. Designing σ_A -targeting curricula remains an open engineering challenge.

Multi-agent collective field dynamics. The present framework treats agent A as the unit of analysis. Full multi-agent systems require extending to a collective knowledge field with emergent schema dynamics.

Parameter calibration. Multiple rate constants ($\rho, \epsilon_\sigma, \gamma, \nu_M, \kappa_P, \kappa_I, \kappa_F$) remain unmeasured. The dimensionless parameter groups reduce independent parameters but do not eliminate empirical estimation needs.

13 Conclusion

The H-Bar Model advances the case that depth, schema coherence, attentional fidelity, executive control, metacognitive self-modelling, and collective schema communication are formally independent variables in agent training — that optimising any subset without the others produces systematically predictable failure modes — and that the field now has both the empirical tools and the formal framework to test this claim rigorously.

Appendix A: Notation Reference

Symbol	Type	Domain	Description
D	Set	Global	Knowledge domains, indexed by $d \in D$
M	Set	Global	Modalities {text, visual, auditory, sensorimotor, symbolic}
$\Delta(d, t)$	Function	$\mathbb{R}_+$	Domain frontier (maximum achievable depth)
$\delta_A(d, t)$	State var	$[0, \Delta(d, t)]$	Parametric depth (structural complexity)
$\delta_A^{\text{relative}}(d, t)$	Derived	$[0, 1]$	Relative depth (fraction of frontier achieved)
$\beta_A(d, t)$	State var	$\mathbb{R}_+$	Functional breadth (surface competence)
$\sigma_A(d, t)$	State var	$[0, 1]$	Schema coherence (principled reorganisation degree)
$\tilde{\sigma}_A(d, t)$	Estimate	$[0, 1]$	Stage 1 training-time proxy (multi-signal fusion)
$\hat{\sigma}_A(d, t)$	Estimate	$[0, 1]$	Stage 2 evaluation-time ground-truth (OOD/ID ratio)
$\alpha_A(d, t)$	State var	$[0, 1]$	Attentional fidelity (structure-directed attention)
$\bar{M}_A(d, t)$	State var	$[0, 1]$	Self-model of schema coherence (metacognitive estimate)
$\zeta_A(d, t)$	Derived	$[-1, 1]$	Calibration error ($\bar{M}_A - \sigma_A$)
$\Xi_A^p(t)$	State var	$[0, 1]$	Executive planning quality
$\Xi_A^i(t)$	State var	$[0, 1]$	Executive inhibition quality
$\Xi_A^f(t)$	State var	$[0, 1]$	Executive cognitive flexibility
$\Psi_A(d_1, d_2, t)$	Derived	$\mathbb{R}_+$	Intersection activation (cross-domain discovery rate)
$q_A(d, t)$	Derived	$[0, 1]$	Effective mastery quality ($\sigma_A \cdot \delta_A^{\text{relative}}$)
$M_A(t)$	Set	2^D	Mastery set (domains with high δ_A and σ_A)
$B_A(t)$	Set	2^D	Breadth set (domains with high β_A but not mastered)
$D^*(d, t)$	Derived	2^d	Delegation gradient (skills safe for AI handoff)
λ_c	Parameter	$\mathbb{R}_+$	Parametric decay rate
λ_c^{eff}	Derived	$\mathbb{R}_+$	Effective decay (schema-mediated reduction)
$\lambda_f(d, t)$	Parameter	$\mathbb{R}_+$	Frontier obsolescence rate
$\eta(d, t)$	Derived	$\mathbb{R}_+$	Learning efficiency (Gompertz form)
$\eta_{\max}$	Parameter	$\mathbb{R}_+$	Maximum learning efficiency
ρ	Parameter	$\mathbb{R}_+$	Schema growth rate constant
ϵ_σ	Parameter	$\mathbb{R}_+$	AI bypass suppression coefficient
$\Omega_{AI}(d, t)$	Function	$\mathbb{R}_+$	AI bypass risk (surface-reward pressure)
$P_A(d, t)$	Function	$\mathbb{R}_+$	Principled structure availability
γ	Parameter	$\mathbb{R}_+$	Attention growth rate constant
ζ_α	Parameter	$\mathbb{R}_+$	Surface-reward attention suppression
$R_A^{\text{surface}}(d, t)$	Function	$\mathbb{R}_+$	Surface-reward pressure
ν_M	Parameter	$\mathbb{R}_+$	Self-model convergence rate
ξ_M	Parameter	$\mathbb{R}_+$	AI bypass metacognitive inflation
$\kappa_P, \kappa_I, \kappa_F$	Parameters	$\mathbb{R}_+$	Executive control convergence rates
$\mu_{AB}(d, t)$	Derived	$[0, 1]$	Schema legibility (inter-agent)
$\Sigma_{A,B}(d_1, d_2, t)$	Derived	$[0, 1]$	Collective schema field
$\tau_A(B, d, t)$	Parameter	$[0, 1]$	Theory-of-mind coupling. (Note: bare τ without subscript denotes the segmented-regression breakpoint time in Prediction 9.)
$\Theta_A(d, m_1, m_2, t)$	Derived	$[0, 1]$	Cross-modal schema transfer
$\omega(m_1, m_2)$	Parameter	$[0, 1]$	Modality similarity weight
$V_A(B, f, t)$	Derived	$[0, 1]$	Benchmark validity function
$CI(B, f)$	Metric	$[0, 1]$	Construct isolation
$FD(B)$	Metric	$[0, 1]$	Format diversity
$DG(B)$	Metric	$[0, 1]$	Difficulty gradient
$R_A(B, f, t)$	Metric	$[0, 1]$	Benchmark reliability
σ_{critical}	Derived	$[0, 1]$	Phase 1→2 transition threshold
R_0	Derived	$\mathbb{R}_+$	Basic schema reproduction number
$\theta_I, \theta_\delta, \theta_\sigma, \theta_\beta$	Parameters	$[0, 1]$	Threshold parameters
γ_σ	Parameter	$[0, 1]$	Schema-mediated decay protection
w_g, w_r, w_c	Parameters	$[0, 1]$	Signal fusion weights (Stage 1)

Visual syntax conventions:

- No decoration (e.g., σ_A): true latent variable
- Tilde (e.g., $\tilde{\sigma}_A$): training-time proxy estimate (Stage 1)
- Hat (e.g., $\hat{\sigma}_A$): evaluation-time ground-truth estimate (Stage 2)
- Superscript “relative” (e.g., $\delta_A^{\text{relative}}$): normalised to $[0, 1]$
- Superscript “eff” (e.g., λ_c^{eff}): effective/derived quantity
- Subscript A : agent-relative quantity
- Subscript AI : AI-system quantity

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